von Neumann stability analysis: FTCS advection.

- Advection equation (moving to the right):

$$\frac{\partial a_-}{\partial t} = -c\frac{\partial a_-}{\partial x}, \text{ with solution } a_- = f(x-ct).$$

- FTCS update equation:

$$a_j^{n+1} = a_j^n - \frac{1}{2}g\left(a_{j+1}^n - a_{j-1}^n\right), \text{ with } g = \frac{c\tilde{\tau}}{h}.$$

- We assume a solution is made of waves:

$$a(x,t) = A(t)\,e^{ikx}, \text{ where } i=\sqrt{-1},\ k \text{ - wave number}$$

$A(t)$ - complex, time-dependent amplitude.

In discrete form: $a_j^n = A^n e^{ikx_j} = A^n e^{ikjh}$
(note: $x_j = jh$).

- Substitute the discrete form to the update equation:

$$A^{n+1}\cdot e^{ikjh} = A^n\cdot e^{ikjh} - \frac{1}{2}g\left(A^n e^{ik(j+1)h} - A^n e^{ik(j-1)h}\right)$$

divide both sides by $A^n e^{ikjh}$

$$\frac{A^{n+1}}{A^n} = 1 - \frac{1}{2}g\left(e^{ikh} - e^{-ikh}\right)$$

But, from Euler's Identity:

$$(e^{ikh} - e^{-ikh}) = 2i\sin(kh).$$

and $\xi = \dfrac{A^{n+1}}{A^{n}}$.

So $\xi = 1 - ig\sin(kh)$

- For the scheme to be stable $|\xi| \leq 1$.

If $|\xi| > 1$, the wave grows exponentially.

Since ξ is a complex number

$$|\xi|^2 = Re(\xi)^2 + Im(\xi)^2$$

$$|\xi|^2 = (1)^2 + (-ig\sin(kh))^2$$

$$|\xi|^2 = 1 + g^2\sin^2(kh)$$

$$|\xi| = \sqrt{1 + g^2\sin^2(kh)}$$

since $g^2 \geq 0$ (no equality for non-zero c, τ)

$\sin^2(kh) \geq 0$

$\Rightarrow$ $\boxed{|\xi| > 1 \text{ for all } \tau \text{ and most } k.}$

NB: $\sin^2(kh) = 0$ for $k = 0$ or $k = \pi/h$. For these k $|\xi| = 1$.

However, for a numerical scheme to be considered stable, it must be stable for every possible wave!

FTCS is unconditionally unstable